# Physics-Derived Features for Future-Time Prediction on Unseen Rooftop PV Systems

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## Abstract

Data-driven photovoltaic (PV) power models can appear accurate when training and test data share installations or timestamps, but new rooftop systems require transfer across both system and time. This study tests whether physically derived, system-independent features improve prediction for unseen PV systems under two evaluation protocols. The analysis uses the HKUST rooftop PV dataset and retains 1,370,191 modeling-ready daylight observations from 37 SolarEdge systems. Weather-only XGBoost is compared with an otherwise identical hybrid model augmented with solar zenith, solar azimuth, clear-sky irradiance, clear-sky index, and estimated cell temperature. In the same-period leave-systems-out evaluation, macro nRMSE falls from 0.0919 to 0.0790, a 14.0% reduction. A stricter future-time experiment trains only on pre-2023 observations from other systems and tests held-out systems during 2023, covering 612,371 test rows. In this setting, a time/solar-only model, weather-only XGBoost, and the hybrid obtain macro nRMSE of 0.1670, 0.1330, and 0.1114, respectively. The hybrid reduces nRMSE by 16.2% relative to weather-only XGBoost (paired bootstrap 95% interval: 14.2%-18.2%) and has lower error on all 37 systems, including four with no eligible pre-2023 history. Absolute error is higher than in the same-period experiment and the hybrid overpredicts 2023 output by 3.36% of capacity on average. The results support physics-derived features for future-time prediction on unseen systems within one campus, conditional on contemporaneous measured weather; they do not establish day-ahead or cross-climate forecasting.

**Keywords:** *photovoltaics; physics-informed machine learning; XGBoost; domain generalization; cold-start prediction; rooftop solar*

## 1. Introduction

Rooftop photovoltaic generation is inherently variable because the amount of electricity produced at a given moment depends on solar geometry, atmospheric attenuation, clouds, temperature, wind, installation capacity, orientation, and equipment condition. This variability complicates monitoring and planning for distributed solar fleets. Prediction models can help estimate expected production, detect abnormal behavior, and support grid and building operations, but their usefulness depends on how they are evaluated. A model that performs well on randomly held-out timestamps from a familiar system may be learning that system's characteristic scale and response rather than a relationship that transfers to a new installation.

The cold-start setting is especially important for new or sparsely monitored rooftop systems. A newly commissioned system has no long history of measured output from which to learn site-specific behavior. In such a setting, a useful model must transfer relationships learned from other installations. Prior research has shown that a single XGBoost model can nowcast individual yield across a large residential fleet and that models trained on geographically aggregated data can generalize to individual systems (Costa, 2022; Grzebyk et al., 2023). More recent work has addressed zero-sample PV prediction using explicit domain-generalization architectures (Liu et al., 2025). These studies establish that cross-system prediction is possible and technically distinct from ordinary within-system forecasting.

Physical knowledge offers one possible route to improved transfer. Measured global horizontal irradiance (GHI) captures current sunlight, but the same GHI value can occur under different solar positions and clear-sky expectations. Solar zenith and azimuth encode geometry; clear-sky GHI provides a time- and location-specific atmospheric reference; the clear-sky index expresses observed irradiance relative to that reference; and an estimated cell temperature represents the thermal conditions that influence PV conversion efficiency. These variables are derived from relationships that apply across systems and therefore may help a model rely less on installation-specific statistical patterns.

This paper evaluates that hypothesis using a controlled feature-ablation experiment. Two XGBoost regressors use the same target, grouped folds, algorithm, hyperparameters, and random seed. Model A receives four measured weather variables. Model B receives the same four variables plus five physics-derived features. Because the only planned treatment difference is the feature set, the error difference between the two models estimates the predictive value of those physics features under the chosen evaluation protocol. XGBoost is appropriate for this comparison because tree boosting captures nonlinear interactions in tabular data and has demonstrated strong performance in large PV fleets (Chen & Guestrin, 2016; Grzebyk et al., 2023).

The study makes four contributions. First, it implements a system-grouped comparison in which all observations from a test system are excluded from training. Second, it adds a stricter future-time protocol in which test systems and test timestamps are both unseen. Third, it compares a time/solar-only diagnostic, weather-only XGBoost, and a physics-enhanced hybrid to determine whether performance can be explained by the typical solar-time curve alone. Fourth, it reports system-level macro metrics and paired uncertainty so installations with more observations do not dominate the conclusion.

The completed same-period analysis reports a 14.0% reduction in macro nRMSE for the hybrid relative to weather-only XGBoost. The future-time analysis is harder in absolute terms, but the hybrid still reduces macro nRMSE from 0.1330 to 0.1114, a 16.2% relative reduction with a paired-bootstrap 95% interval of 14.2%-18.2%. It improves all 37 systems, while a time/solar-only model remains substantially less accurate at 0.1670. All systems nevertheless share one campus, climate, and weather station, and the future models receive contemporaneous measured weather. The result therefore supports bounded future-time, cross-system prediction rather than day-ahead or cross-climate forecasting.

## 2. Problem Definition and Hypotheses

The prediction unit is one daylight observation for one rooftop PV system. For system s at timestamp t, the response variable is the measured AC power P divided by the system's installed capacity C. Capacity normalization places systems of different sizes on a comparable scale:

$$y(s,t) = P(s,t) / C(s)$$

The normalized target is approximately interpretable as a fraction of installed capacity. It is not a capacity factor averaged over a long period; it is an instantaneous normalized power value. Rows outside the accepted range from -0.001 to 1.2 are marked impossible and excluded from modeling. The small negative tolerance accommodates numerical or measurement noise, while the upper bound allows modest output above nominal capacity without admitting extreme values.

The study uses two nested evaluation conditions. Phase 5 tests zero historical power data from a held-out system while allowing training and test systems to share timestamps. Phase 6 retains the same system isolation and additionally requires every test timestamp to occur after every training timestamp. For each Phase 6 fold, the regressor is fitted using modeling-ready rows through December 31, 2022 from the non-held-out systems and is evaluated on 2023 rows from the held-out systems. This design distinguishes same-period cross-system transfer from future-time prediction on an unseen system.

Two feature sets define the controlled comparison. Model A uses measured GHI, ambient temperature, wind speed, and relative humidity. Model B uses those same variables plus apparent solar zenith, solar azimuth, clear-sky GHI, clear-sky index, and a GHI-based Faiman cell-temperature estimate. The null and alternative hypotheses are:

- H0: Adding the five physics-derived features does not reduce mean system-level nRMSE relative to the weather-only XGBoost model.
- H1: Adding the five physics-derived features reduces mean system-level nRMSE relative to the weather-only XGBoost model.
- H2: The physics-enhanced model retains lower system-level nRMSE when both the held-out systems and their target timestamps are absent from training.

The primary outcome is system-level nRMSE averaged equally across held-out systems. Secondary outcomes are normalized mean absolute error (MAE) and mean bias error (MBE). Models are compared in paired form because they use the same systems, folds, timestamps, and eligible rows. Phase 5 estimates same-period cross-system transfer; Phase 6 estimates future-time transfer to unseen systems conditional on contemporaneous measured weather. Neither protocol forecasts future weather or evaluates operational day-ahead power.

# 3. Related Research

## 3.1 Data-driven PV prediction

Machine-learning models are attractive for PV prediction because the mapping from meteorological inputs to power is nonlinear and installation-dependent. XGBoost constructs an additive ensemble of decision trees and includes regularization and subsampling mechanisms that support scalable learning from large tabular datasets (Chen & Guestrin, 2016). In a study of 1,102 residential systems, Grzebyk et al. (2023) selected XGBoost for individual yield nowcasting because of its performance and practical simplicity. Their results demonstrated that one fleet-level model can outperform commercial software for individual systems, supporting the use of a shared learner rather than a separate model per installation.

Deep sequence models provide another path. Costa (2022) compared LSTM, convolutional, and hybrid convolutional-LSTM networks for household PV generation and evaluated whether models trained on regionally aggregated data could predict individual systems. The reported results supported regional-to-individual transfer. That study is relevant because it treats generalization as an explicit evaluation question, rather than assuming that accuracy on randomly held-out observations implies transfer.

## 3.2 Physics-based and physics-derived representations

A physics-based PV workflow typically transforms time and location into solar position, estimates clear-sky or actual irradiance components, estimates module temperature, and then maps irradiance and temperature to electrical output. The open-source pvlib library provides modular implementations of solar-position, irradiance, thermal, and electrical models, allowing researchers to use physical calculations either as complete models or as components of hybrid pipelines (Anderson et al., 2023). This study uses pvlib to construct physically meaningful inputs while leaving the final weather-to-power mapping to XGBoost.

The clear-sky reference uses the Ineichen-Perez formulation. Ineichen and Perez (2002) developed an airmass-independent formulation of the Linke turbidity coefficient and associated clear-sky irradiance models. Dividing observed GHI by modeled clear-sky GHI produces a clear-sky index that partially separates predictable solar geometry from cloud-related attenuation. This representation can make the meaning of a given irradiance value more comparable across hours and seasons.

Temperature is another important physical pathway because PV conversion efficiency changes with cell temperature. The Faiman model estimates module or cell temperature using incident irradiance, ambient temperature, wind speed, and empirical heat-loss coefficients (Faiman, 2008). The current implementation applies the default coefficients u0 = 25.0 and u1 = 6.84. However, it substitutes GHI for plane-of-array irradiance because the system-level records may aggregate arrays with multiple or uncertain orientations. This approximation creates a useful thermal proxy but should not be interpreted as a fully specified module temperature model.

### 3.3 Generalization to unseen PV systems

The literature increasingly distinguishes interpolation within familiar sites from prediction on unknown systems. Liu et al. (2025) proposed a generative adversarial domain-generalization network for zero-sample prediction and evaluated nine geographically and capacity-diverse PV systems. Their results show that unseen-system prediction is now an active research area rather than an entirely untested problem. The present study therefore does not claim to be the first cross-system PV model. Its contribution is a transparent controlled ablation on a larger set of installations within one campus, designed to quantify the incremental value of a compact physics feature set in a standard XGBoost pipeline.

This distinction matters for novelty. Complex domain-generalization models may maximize predictive accuracy, but they combine architecture, augmentation, and representation changes. A simpler paired comparison can answer a different question: how much improvement is obtained by adding five physically interpretable variables while holding the learner fixed? A positive answer would support physics-derived feature engineering as a low-complexity baseline for future domain-generalization research. A negative answer would be equally informative because it would show that physical transformations do not automatically improve transfer.

### 3.4 Research gap

Existing studies demonstrate fleet-level prediction, regional generalization, and advanced zero-sample domain generalization, but they do not determine the value of this exact physics feature set under a system-grouped ablation on the HKUST rooftop dataset. The research gap is therefore dataset- and design-specific: the incremental cross-system benefit of solar position, clear-sky context, and a thermal proxy has not been quantified for these campus rooftops under an otherwise identical XGBoost configuration. The study addresses that bounded gap and avoids a broader priority claim that the literature cannot support.

## 4. Dataset and Data Preparation

### 4.1 Source dataset and analytical subset

Lin et al. (2025) released a three-year rooftop PV dataset collected at the Hong Kong University of Science and Technology. The source contains inverter-level generation measurements from 60 gridconnected rooftop stations at five-minute resolution, on-site meteorological measurements at one-minute resolution, and system metadata. The campus is located in Hong Kong's humid subtropical climate. The source paper explicitly notes that the single-location climate may limit the generalizability of models trained on the data.

The current project uses 37 SolarEdge systems out of the 60 stations in the dataset and combines their records into a table with 2,756,516 rows aligned to 15-minute timestamps. This optimizer-equipped SolarEdge cohort provides a consistent hardware scope for the controlled cross-system analysis; the paper makes no claims about stations outside this cohort.

**Table 1. Dataset scope after preprocessing**

| Quantity | Value | Interpretation |
| --- | --- | --- |
| Source period | 2021-2023 | Three years of campus operation |
| Source stations | 60 | All rooftop stations described by Lin et al. (2025) |
| Analyzed systems | 37 | SolarEdge systems in the combined modeling table |
| Raw combined rows | 2,756,516 | Before daylight filtering |
| Daylight rows | 1,387,682 | Apparent zenith below 90 degrees |
| Modeling-ready rows | 1,370,191 | After exclusion-quality flags |
| Flagged daylight rows | 17,491 | Retained for audit but excluded from modeling |

*Source: Study analysis outputs; source-dataset description from Lin et al. (2025).*

## 4.2 Timestamp handling and daylight filtering

Timestamps are localized to the Asia/Hong_Kong time zone rather than converted from another zone. Solar position is calculated for the campus coordinates 22.3364 degrees north, 114.2654 degrees east, at an assumed altitude of 60 m. Apparent solar zenith and azimuth are merged back to all system rows by timestamp. Only observations with apparent zenith below 90 degrees are retained. Removing nighttime rows avoids a misleading reduction in error from long sequences in which both observed and predicted power are trivially zero.

The daylight filter retained 1,387,682 of 2,756,516 rows. The maximum retained zenith was 89.9997 degrees. Because the same campus weather station is shared across systems, the analysis also verifies that weather columns do not vary among stations at a common timestamp. This many-system/one-weather-source structure is useful for isolating system differences but limits claims about geographic transfer.

## 4.3 Physics feature construction

Clear-sky GHI is computed for every retained timestamp with pvlib's Ineichen model. The clear-sky index is the measured GHI divided by the corresponding clear-sky GHI and is capped at 2.0 by the library function. A GHI-based cell-temperature proxy is computed with the Faiman model using ambient temperature, wind speed, u0 = 25.0, and u1 = 6.84. Finally, normalized power is calculated as measured power divided by the summed installed capacity for the system. The engineered values are merged by timestamp or computed row by row, without using target-system power information to construct the weather or solar-geometry inputs.

## 4.4 Quality control

The preprocessing pipeline is non-destructive: quality-control flags remain in the cleaned table, while the modeling-ready indicator determines which rows enter training and evaluation. A daytime fault zero is defined as zero power when GHI is at least 200 W/m2. A stuck measurement is an exactly repeated positive power or weather value lasting at least 16 consecutive 15-minute samples, equivalent to four hours. Additional rules flag missing model variables, nonpositive capacity, negative GHI or wind speed, relative humidity outside 0-100%, and normalized power outside -0.001 to 1.2.

The executed quality summary identified 16,991 rows with missing values and 500 daytime fault-zero rows. It detected no impossible values and no four-hour stuck power or weather runs. Seven hundred thirty-one rows occurred after a detected data gap, but this informational flag did not exclude rows because the tabular models do not use lagged features. In total, 17,491 daylight rows were excluded from modeling. The exact thresholds are transparent and reproducible, but they remain analyst choices; sensitivity to the fault-zero and stuck-sensor thresholds is a potential robustness check.

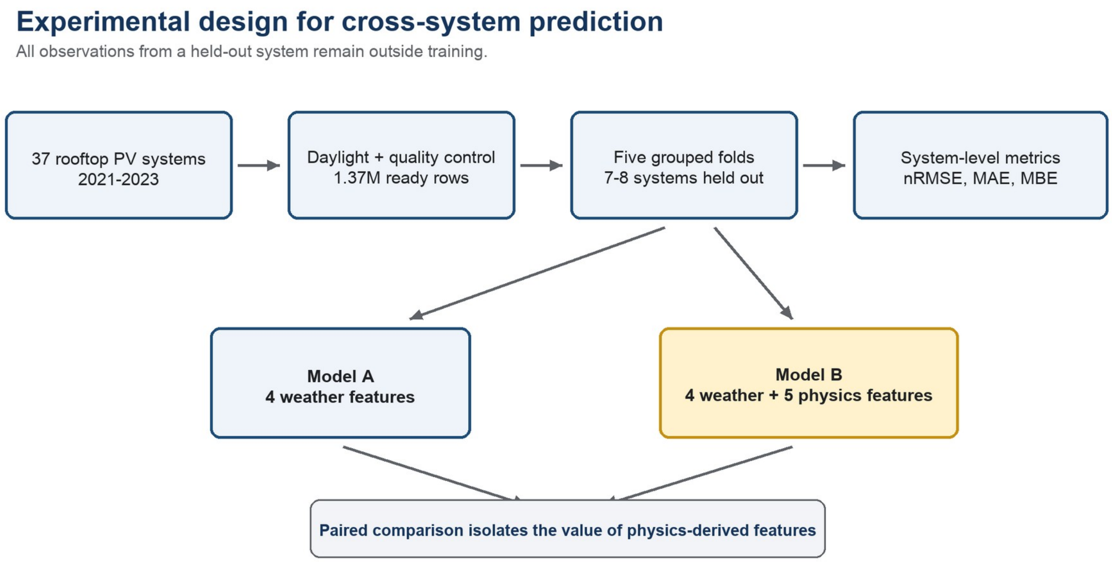


**Figure 1. Same-period leave-systems-out experimental design. All timestamps from each held-out PV system remain outside the training set. Models A and B use identical folds and XGBoost settings, so their paired error difference isolates the added physics-derived features. Phase 6 adds the non-overlapping time boundary described in Section 5.8. Source: Study design.**

# 5. Methodology

## 5.1 Feature-ablation design

The principal experiment is a controlled feature ablation. Model A and Model B use the same regression algorithm, objective, hyperparameters, capacity-normalized target, model-ready rows, grouped folds, and random seed. Model A contains four measured weather features. Model B adds five deterministic physics-derived features. Plane-of-array irradiance is deliberately excluded because a single system-level record may combine arrays with different tilts or azimuths, making one orientation estimate potentially misleading.

**Table 2. Controlled feature sets**

| Feature | Model A | Model B | Rationale |
|---|---|---|---|
| Measured GHI | Yes | Yes | Current solar resource |
| Ambient temperature | Yes | Yes | Environmental and thermal context |
| Wind speed | Yes | Yes | Convective cooling context |
| Relative humidity | Yes | Yes | Atmospheric context |
| Apparent zenith | No | Yes | Solar elevation geometry |

| Feature | Model A | Model B | Rationale |
|---|---|---|---|
| Solar azimuth | No | Yes | Solar direction geometry |
| Clear-sky GHI | No | Yes | Time/location-specific irradiance reference |
| Clear-sky index | No | Yes | Observed irradiance relative to clear sky |
| GHI-based cell temperature | No | Yes | Approximate thermal state |

*Source: Feature definitions used in the study analysis.*

## 5.2 Reference baselines

The training-mean baseline predicts the mean normalized power of the training rows in each fold. It is intentionally weak but establishes the error of a constant predictor under the same split. The GHI physics baseline first computes a solar signal equal to clear-sky GHI divided by 1,000 and multiplied by the clear-sky index. Because the index reconstructs observed GHI up to the clipping rule, the signal is a capacity-normalized irradiance proxy. A nonnegative scalar is fit through the origin using only training rows, and predictions are clipped between 0 and 1.2. The fitted scalar ranges from 0.6254 to 0.6382 across the five folds.

This baseline is not a full PVWatts or system-specific physical model. It does not use system tilt, azimuth, module type, inverter efficiency, or loss parameters. Its purpose is to test how far a simple physical irradiance relationship can go without target-system calibration.

## 5.3 XGBoost specification

Both machine-learning models use XGBoost regression with squared-error loss. Gradient-boosted trees sequentially add trees that correct residual error while regularization and subsampling reduce overfitting (Chen & Guestrin, 2016). Tree models do not require feature standardization, so scaling is disabled. No separate hyperparameter tuning is performed for Models A and B; tuning them independently would confound the feature comparison.

**Table 3. Shared XGBoost configuration**

| Parameter | Value | Purpose |
|---|---|---|
| objective | reg:squarederror | Squared-error regression |
| n_estimators | 1,000 | Number of boosting rounds |
| learning_rate | 0.05 | Shrinkage per tree |
| max_depth | 6 | Maximum interaction depth |
| min_child_weight | 10 | Minimum child-node weight |
| subsample | 0.8 | Row subsampling |

| Parameter | Value | Purpose |
| --- | --- | --- |
| colsample_bytree | 0.8 | Feature subsampling per tree |
| reg_alpha | 0.0 | L1 regularization |
| reg_lambda | 1.0 | L2 regularization |
| tree_method | hist | Histogram-based training |
| random_state | 42 | Deterministic split/model seed |

*Source: Model configuration used in the study analysis.*

## 5.4 Leave-systems-out evaluation

Five-fold GroupKFold cross-validation is used with system ID as the group variable. The random seed is 42 and shuffling is enabled. Each fold contains approximately 29-30 training systems and 7-8 test systems. Every system appears in a test fold exactly once. Explicit assertions verify that the training-system and test-system sets are disjoint and that each expected system receives one evaluation result. This design follows the grouped-validation principle implemented in scikit-learn (Pedregosa et al., 2011).

Predictions are evaluated at two levels. Fold-level metrics summarize all rows in a fold, while system-level metrics are calculated separately for each held-out installation. Headline values are macro means across the 37 system results, giving each system equal weight regardless of its number of observations. This choice aligns the estimand with the question: expected performance on an unseen system, not expected error on a randomly selected timestamp from the pooled dataset.

## 5.5 Performance metrics

Because the target is already divided by installed capacity, all errors are expressed as fractions of capacity. For observed normalized power y_i and prediction y-hat_i over N observations for a system, the metrics are:

$$nRMSE = \sqrt{[(1/N)\ \Sigma_i\ (\hat{y}_i - y_i)^2]}$$

$$MAE = (1/N)\ \Sigma_i\ |\hat{y}_i - y_i|$$

$$MBE = (1/N)\ \Sigma_i\ (\hat{y}_i - y_i)$$

Lower nRMSE and MAE indicate greater accuracy. MBE is signed: positive values indicate average overprediction and negative values indicate average underprediction. The relative error reduction of Model B against Model A is 100 times the difference between Model A and Model B error divided by Model A error.

## 5.6 Leakage controls and reproducibility

The analysis pipeline contains several safeguards. All rows from a system remain in one fold, and preprocessing that can learn from data is fitted on training rows only. Models A and B use identical group assignments. The cleaned table retains time-zone information and verifies unique system-timestamp rows. Model-ready rows contain no missing model variables or active exclusion flags. Phase 6 adds explicit assertions that training and test systems are disjoint, that the latest training timestamp precedes the earliest test timestamp, and that no timestamp appears in both partitions. Fixed random seeds support deterministic reruns.

Reproducibility is improved: the Phase 6 notebook uses a portable experiment-relative data path, and its fold, system, summary, paired-bootstrap, and figure outputs are preserved together in the project. Dependency versions remain unpinned, and the complete pipeline has not yet been reproduced from a clean environment. Before submission, dataset retrieval and software versions should be documented and a clean rerun should confirm every saved value.

## 5.7 Paired statistical comparison

The uncertainty analysis is paired at the system level. For each held-out system, the nRMSE difference is computed as Model B minus Model A. A deterministic paired bootstrap resamples the 37 systems with replacement 10,000 times using seed 42, recalculates the macro difference and relative reduction, and uses the 2.5th and 97.5th percentiles as a 95% interval. Resampling rows would understate uncertainty because adjacent timestamps from the same system are dependent. The executed comparison gives a mean nRMSE difference of -0.0129, with a 95% interval from -0.0146 to -0.0109; the corresponding relative reduction is 14.0%, with a 95% interval from 11.3% to 16.5%. Model B has lower nRMSE on 36 of 37 systems, with no ties.

## 5.8 Future-time prediction on unseen systems

Phase 6 assigns every system to one of five deterministic held-out folds. Training uses only modeling-ready observations through December 31, 2022 from the non-held-out systems, while testing uses January 1 through December 31, 2023 observations from the held-out systems. Each fold contains seven or eight test systems and 117,255 to 133,631 test rows. Because four systems begin in 2023, the actual number of systems contributing historical training rows ranges from 26 to 28. Across folds, every one of the 37 systems is tested exactly once, totaling 612,371 future observations.

The future-time suite contains a training-mean baseline, a time/solar-only XGBoost model using solar zenith, azimuth, and clear-sky GHI, the four-feature weather-only Model A, and the nine-feature hybrid Model B. The time/solar-only diagnostic tests whether a typical daily and seasonal output curve can explain performance without measured weather. The three XGBoost models use the same configuration, and all four models use identical station folds and eligible test rows.

Phase 6 uncertainty is calculated with the same system-level paired bootstrap principle as Phase 5. Ten thousand resamples of the 37 systems are drawn with replacement using seed 42. For nRMSE, the reported interval is computed for both the absolute hybrid-minus-weather difference and the relative reduction. Timestamp rows are not resampled as independent observations.

# 6. Experimental Results

## 6.1 Physics baseline substantially improves on a constant predictor

The calibrated GHI baseline reduced macro nRMSE from 0.2160 to 0.1237, a relative reduction of 42.7%. It reduced macro MAE from 0.1811 to 0.0776, or 57.2%. The GHI baseline had lower nRMSE than the training-mean predictor on all 37 held-out systems. This result is a useful validation of the experimental setup: a simple irradiance-based relationship contains transferable information that the constant baseline lacks.

The physics baseline nevertheless underpredicted on average, with macro MBE of -0.0170. It also remained materially less accurate than either XGBoost model. Therefore, physics alone provides a meaningful reference but does not capture all nonlinear relationships between weather and PV output in the analyzed systems.

## 6.2 Physics features reduce average XGBoost error by approximately 14%

The completed Phase 5 analysis reports macro nRMSE of 0.0919 for the weather-only model and 0.0790 for the physics-enhanced model. The absolute difference is -0.0129 of installed capacity, corresponding to a 14.0% relative reduction. Macro MAE falls from 0.0614 to 0.0522, a 15.0% relative reduction. Mean bias also moves closer to zero, from 0.0010 for Model A to -0.0004 for Model B.

**Table 4. Macro performance across 37 held-out PV systems**

| Model | Macro nRMSE | SD | Macro MAE | Macro MBE |
|---|---|---|---|---|
| Training mean | 0.2160 | 0.0185 | 0.1811 | -0.0013 |
| Calibrated GHI physics | 0.1237 | 0.0146 | 0.0776 | -0.0170 |
| Model A: weather-only XGBoost | 0.0919 | 0.0190 | 0.0614 | 0.0010 |
| Model B: physics-enhanced XGBoost | 0.0790 | 0.0228 | 0.0522 | -0.0004 |

*Source: Study analysis outputs.*

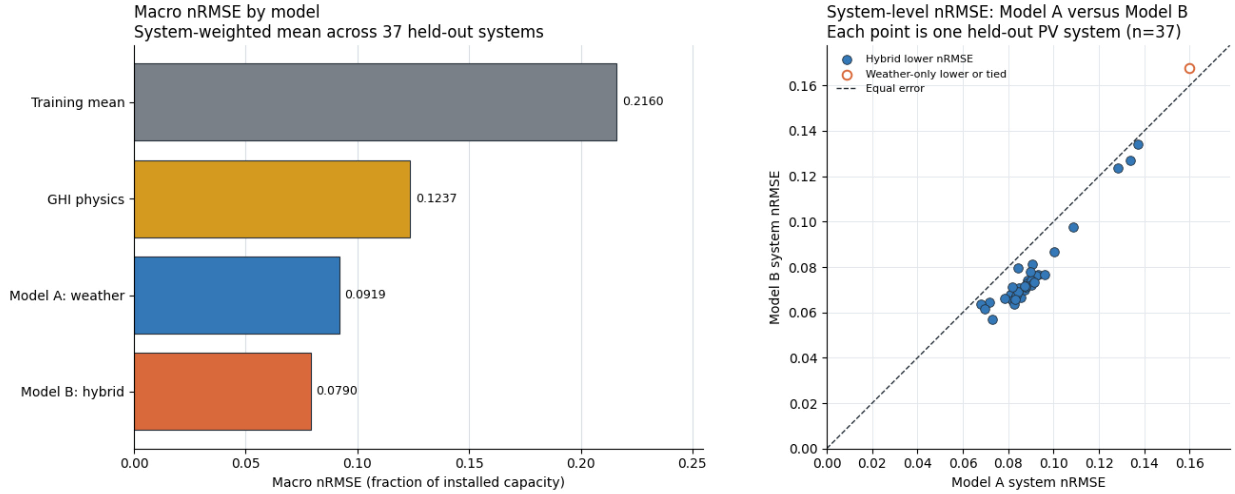


**Figure 2. Phase 5 model comparison. The left panel shows macro nRMSE across the four models. The right panel compares weather-only Model A with physics-enhanced Model B for each held-out PV system; points below the identity line favor Model B. Model B has lower nRMSE on 36 of 37 systems. Source: Study analysis outputs.**

## 6.3 The gain is broad but not universal across systems

Aggregate improvement is not the only relevant result. Model B's system-level nRMSE standard deviation is 0.0228, compared with 0.0190 for Model A. Its median is lower, 0.0720 versus 0.0879, suggesting that a typical system benefits. However, the maximum system nRMSE is slightly worse for Model B, 0.1676 versus 0.1598. These values imply heterogeneous effects: physics features improve the macro average and median but may harm at least one difficult system.

This heterogeneity is scientifically important because the practical question concerns reliability on a new system, not only average accuracy. The paired Phase 5 comparison shows that Model B has lower nRMSE on 36 of 37 systems, with no ties. The scatterplot in Figure 2 places nearly every system below the identity line, indicating that the macro improvement is broad rather than driven by a small subset. One system nevertheless favors Model A, consistent with Model B's slightly worse maximum system error.

## 6.4 Paired uncertainty supports the within-campus improvement

The observed 14.0% reduction is a point estimate over 37 systems. In the 10,000-resample paired bootstrap, the 95% interval for the relative nRMSE reduction is 11.3%-16.5%, and the interval for the absolute Model B minus Model A difference is -0.0146 to -0.0109. Both intervals favor Model B. Because the resampling unit is the held-out PV system, this result supports a stable improvement across systems represented by the campus sample. It does not establish causality or guarantee transfer to different climates, hardware, or data-generating processes.

## 6.5 Interpretation analysis

No feature-importance analysis is available to determine whether the hybrid model relies primarily on clear-sky index, solar geometry, or the temperature proxy. This analysis is not required to establish the main feature-ablation result, but it would strengthen the explanation of why Model B improves.

## 6.6 Temporal separation raises absolute error but preserves the hybrid advantage

The station-and-time-disjoint experiment is materially harder than the same-period evaluation. Weather-only macro nRMSE rises from 0.0919 to 0.1330, a 44.7% increase, while hybrid nRMSE rises from 0.0790 to 0.1114, a 41.0% increase. The overlapping-time protocol therefore understates future-year error, but both experiments retain the same model ordering and a substantial hybrid advantage.

**Table 5. Future-time performance across 37 held-out PV systems**

| Model | Macro nRMSE | SD | Macro MAE | Macro MBE |
|---|---|---|---|---|
| Training mean | 0.2095 | 0.0165 | 0.1780 | 0.0133 |
| Time/solar-only XGBoost | 0.1670 | 0.0082 | 0.1193 | 0.0116 |
| Model A: weather-only XGBoost | 0.1330 | 0.0242 | 0.0986 | 0.0670 |
| Model B: physics-enhanced XGBoost | 0.1114 | 0.0176 | 0.0790 | 0.0336 |

*Source: Phase 6 summary and system-level results. Training uses pre-2023 rows from other systems; testing uses 2023 rows from held-out systems.*

The diagnostic models separate a typical solar-time curve from measured weather and physics context. Time/solar-only XGBoost lowers macro nRMSE from 0.2095 for the training mean to 0.1670, a 20.3% reduction. Weather-only XGBoost reaches 0.1330 and outperforms the time/solar-only model on 32 of 37 systems. The hybrid reaches 0.1114, a 33.3% reduction relative to time/solar-only, and outperforms it on all 37 systems. This pattern shows that recurring solar geometry matters but does not explain the hybrid result.

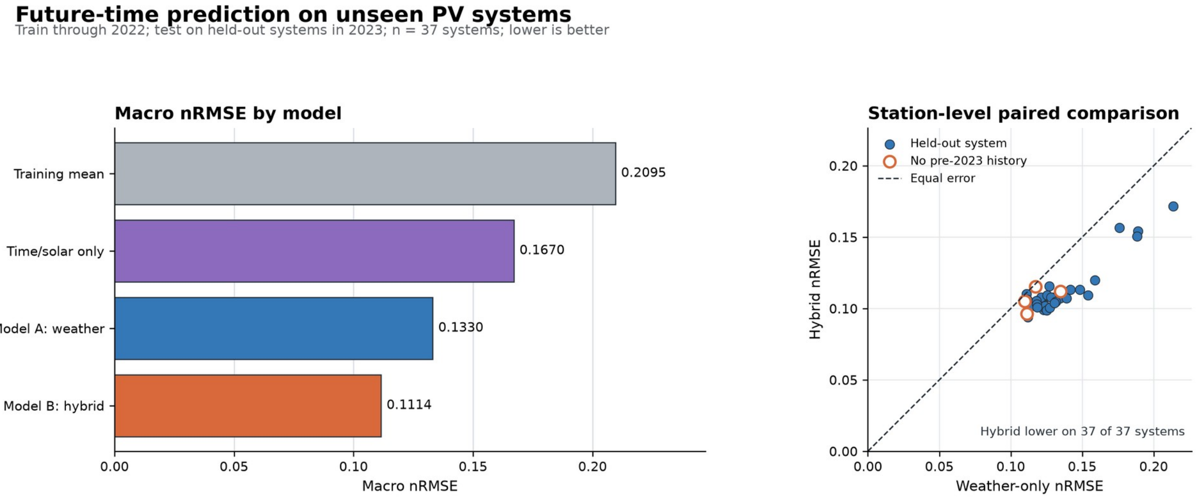


Figure 3. Phase 6 future-time comparison. The left panel shows macro nRMSE when training ends in 2022 and testing uses 2023 observations from held-out systems. The right panel compares weather-only and hybrid nRMSE for each system; all 37 points favor the hybrid. Open circles identify four systems with no eligible pre-2023 history. Source: Phase 6 system-level results.

## 6.7 Physics features improve every future-time held-out system

The system-level paired result is unusually consistent. Hybrid nRMSE is lower than weather-only nRMSE on all 37 systems, with no ties. The mean absolute difference is -0.0216 of installed capacity (95% interval: -0.0251 to -0.0182), and the 16.2% relative reduction has a 95% interval of 14.2%-18.2%. Macro MAE falls from 0.0986 to 0.0790, a 19.9% reduction with a paired-bootstrap interval of 18.0%-21.6%. All five folds favor the hybrid.

The four systems with no eligible pre-2023 history also favor the hybrid. Relative nRMSE reductions are 16.8% for SQ Block P, 13.5% for Shaw Auditorium, 4.3% for UG Hall 8, and 1.8% for UG Hall 9. Their average hybrid nRMSE is 0.1072 versus 0.1183 for weather-only XGBoost, a 9.4% reduction. This subgroup provides direct cold-start evidence, although its smaller and more variable benefit should not be generalized beyond four systems.

Bias remains the main weakness. Weather-only XGBoost overpredicts all 37 systems and has macro MBE of 0.0670. The hybrid reduces absolute bias on every system and lowers macro MBE to 0.0336, but still overpredicts on 35 systems. Thus, the physics features improve both error and calibration without fully correcting the shift between pre-2023 training data and 2023 outcomes.

# 7. Discussion

## 7.1 Why physics-derived features may improve transfer

The results are consistent with the idea that physical transformations provide a more stable representation across systems and years. Raw GHI indicates the instantaneous resource, but it does not directly state whether that irradiance is high or low relative to the current solar position. Clear-sky GHI and clear-sky index add that context. Solar zenith and azimuth encode deterministic daily and seasonal geometry, while the cell-temperature proxy combines irradiance, ambient temperature, and wind in a form more closely related to PV conversion.

The time/solar-only diagnostic clarifies that the hybrid is not merely looking up a typical percentage for a timestamp. Solar geometry alone improves macro nRMSE from 0.2095 to 0.1670, so the daily and seasonal curve is useful. However, the full hybrid reaches 0.1114 and improves every held-out system despite all 2023 timestamps being absent from training. The five-feature bundle therefore contributes transferable weather and physical context beyond a recurring solar-time pattern, although grouped ablation is still needed to isolate each feature's mechanism.

## 7.2 Comparison with related work

The results align with prior evidence that shared models can predict individual PV systems (Costa, 2022; Grzebyk et al., 2023). They also complement the domain-generalization approach of Liu et al. (2025). Whereas that work proposes a complex adversarial framework across nine diverse systems, this study uses a standard learner and 37 systems from one campus to isolate feature representation. The approaches are not direct competitors because their datasets, inputs, prediction horizons, and metrics differ. The contribution here is interpretability and experimental control rather than architectural novelty.

The comparison with the GHI baseline also clarifies the role of hybrid modeling. A simple irradiance-to-power relationship transfers better than a constant predictor, but both XGBoost models improve substantially on it. This suggests that the data-driven learner captures nonlinearities and operational differences not represented by the single physics scale. The hybrid model then improves further, indicating that statistical flexibility and physical context are complementary in this setting.

### 7.3 Practical meaning of the reported error

In the stricter Phase 6 experiment, an nRMSE of 0.1114 means that root-mean-square prediction error is approximately 11.14% of installed capacity for the average held-out system. For a 100 kW system, this corresponds to roughly 11.1 kW RMSE under eligible daylight conditions. Macro MAE is 0.0790, or about 7.9 kW for a 100 kW system, and MBE of 0.0336 corresponds to average overprediction of approximately 3.36 kW. System-specific errors can be substantially higher or lower.

That interpretation should not be confused with day-ahead operational forecasting. Phase 6 predicts later timestamps, but the models receive measured contemporaneous irradiance and weather at those timestamps. Replacing those inputs with weather forecasts would introduce additional error and require a defined forecast horizon. The present model is most directly relevant to conditional prediction, performance benchmarking, and fault-screening applications.

### 7.4 Same-period evaluation is optimistic but directionally reliable

Phase 5 answers whether a model transfers across systems when other systems supply training observations from the same date range. That question remains useful for fleet-level estimation and historical benchmarking, but it is easier than predicting a later year. Phase 6 shows that temporal overlap lowers absolute error by roughly 41%-45%. However, the hybrid advantage persists and grows slightly in relative terms, from 14.0% to 16.2%. The earlier result was therefore optimistic about absolute accuracy but directionally reliable about the value of physics-derived features.

### 7.5 Why future-time error and bias remain system-dependent

Although the hybrid improves nRMSE on all 37 systems in Phase 6, absolute error remains heterogeneous. UG Hall 2 2F, Zone A2, S H Ho Sports Hall, and Zone J2 have the largest hybrid nRMSE, and each shows substantial positive bias. Rooftops differ in orientation, shading, module technology, inverter behavior, commissioning date, degradation, and maintenance. Common clear-sky and temperature proxies cannot represent all of these system-specific factors.

System-level diagnostics should compare error and bias against available metadata such as capacity, optimizer type, orientation complexity, commissioning date, and missing-data rate. The hybrid reduces absolute bias relative to weather-only XGBoost on all 37 systems, but it still overpredicts on 35. A training-period calibration that is evaluated without using target-system history, or richer transferable metadata, may reduce this temporal bias.

## 8. Limitations, Uncertainty, and Threats to Validity

Several limitations bound the conclusions. First, all analyzed systems share one campus, one climate, and one weather station. Holding out systems tests variation in equipment and rooftop configuration, but it does not test geographic, meteorological, or sensor-domain shift. The experiment supports cross-system generalization within the HKUST environment, not cross-climate generalization.

Second, both evaluations are conditional power prediction rather than complete operational forecasts. Phase 6 moves the target timestamps into a later year, but contemporaneous measured weather is still supplied to the models. Numerical weather-prediction inputs would add forecast error and require an explicit horizon. Future-time conditional prediction is therefore more accurate terminology than day-ahead forecasting.

Third, physical feature accuracy is constrained by metadata. Plane-of-array irradiance is omitted because the site-level power series can aggregate arrays with uncertain or mixed orientations. The Faiman temperature proxy therefore uses GHI, and its default coefficients were originally derived for open-rack modules under different conditions (Faiman, 2008). These choices favor robustness and availability over system-specific physical fidelity.

Fourth, the 15-minute rows within a system are temporally dependent. Reporting millions of rows does not imply millions of independent observations. System-level macro evaluation and paired system resampling partially address this issue, but the effective sample size for generalization is closer to 37 systems than to 1.37 million timestamps. Claims of statistical certainty must reflect that grain.

Fifth, temporal robustness is evaluated with one cutoff and one future calendar year. Weather regimes, degradation, and system availability may differ in other years. Four systems have no eligible pre-2023 history and provide useful cold-start evidence, but their average hybrid improvement is smaller than for the remaining systems. Rolling cutoffs and additional years are needed to estimate year-to-year variability.

Sixth, the reported population is restricted to modeling-ready daylight observations. Quality-control rules exclude missing values, physically impossible ranges, stuck measurements, and likely daytime fault zeros; some exclusions therefore use observed power. The reported errors describe expected production during retained operation and should not be interpreted as performance for fault or outage detection.

Seventh, Phase 6 result tables and figures are archived with the experiment, but package versions remain unpinned and a clean end-to-end rerun has not been documented. The saved outputs are internally consistent and reproducible from the station-level CSV, yet final submission should record the execution environment and dataset retrieval procedure.

Finally, the model configuration was selected before the paired comparison but not established through a nested tuning procedure. Holding the configuration fixed is appropriate for isolating feature value, yet it does not show that either feature set is optimally tuned. The result should be interpreted as the improvement under one reasonable common XGBoost configuration.

## 9. Future Work

The first priority is to move from contemporaneous measured weather to operationally available weather forecasts and to repeat the station-and-time-disjoint experiment with rolling temporal cutoffs. This would establish explicit forecast horizons, quantify weather-forecast error, and show whether the 2023 result persists across different years.

A second priority is robustness across hardware and location. The grouped and future-time evaluations should be repeated in an independent hardware population and on a public dataset from a different climate, such as Alice Springs. Such tests would show whether the present within-campus result extends across equipment, weather sensors, and climate domains.

A third direction is mechanism-focused ablation and calibration. Adding solar geometry, clear-sky normalization, and temperature features in groups would identify which representation transfers across time. Calibration methods should also be evaluated using training-system data only to address the positive 2023 bias without introducing target-system history.

Finally, physics-as-features can be compared with physics-guided residual correction. In the latter design, a physical model produces an initial prediction and machine learning models its residual error. Comparing these strategies under the same station-and-time-disjoint folds would test whether physical structure is more transferable as an input representation or as a baseline model.

## 10. Overall Conclusion

This study asks whether a compact set of physics-derived variables improves PV power prediction for rooftop systems that are absent from training, including at future timestamps. In the same-period leave-systems-out evaluation, weather-only and hybrid XGBoost achieve macro nRMSE of 0.0919 and 0.0790. In the stricter future-time experiment, training ends in 2022 and held-out systems are tested during 2023; macro nRMSE is 0.1330 for weather-only XGBoost and 0.1114 for the hybrid.

The Phase 6 difference corresponds to a 16.2% reduction in macro nRMSE and a 19.9% reduction in macro MAE. The paired bootstrap places the relative nRMSE improvement between 14.2% and 18.2%, and the hybrid lowers nRMSE on all 37 held-out systems, including four with no eligible pre-2023 history. The time/solar-only model remains materially less accurate, showing that the result cannot be explained by the typical daily and seasonal output curve alone.

The most defensible conclusion is bounded but stronger than the same-period result alone: physics-derived features improve future-time prediction on unseen rooftop PV systems in this specific campus dataset and controlled XGBoost pipeline. Absolute error rises under temporal separation and the hybrid retains positive 2023 bias. Because contemporaneous measured weather is supplied and all systems share one climate and weather station, the study does not establish day-ahead or cross-climate forecasting.

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